

Md. Rubaet Kabir Zishan

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🌐 Portfolio

🐙 Github

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EDUCATION

B.Sc. in Computer Science & Engineering

May 2022 – May 2026

Military Institute of Science and Technology (MIST), Dhaka, Bangladesh

- **Cumulative GPA:** 3.27 / 4.00
- **Thesis:** *Identification and Translation of Transliterated Chakma Text into Bangla* — bidirectional seq2seq fine-tuning for two linguistically distinct, extremely low-resource languages with minimal parallel corpora. **Successfully defended**; manuscript in preparation for Q1/Q2 journal submission.
- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Algorithms & Complexity, Software Engineering, Computer Networks, Database Systems, Probability & Statistics.

RESEARCH INTERESTS

Applied Machine Learning

Explainable AI (XAI)

Natural Language Processing

AI Security & Cybersecurity

Large Language Models

Reinforcement Learning

Probabilistic Modeling

Agentic & Multi-Agent AI

AI for Environment & Health

Low-Resource Language Tech

PUBLICATIONS

“**AI Integrated Automated Door Lock System Through Face Recognition.**” *IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, 2025. [Published — IEEE Xplore]

“**Chashi-IoT: Developing a Sustainable Integrated Agricultural IoT System Using Cisco Packet Tracer.**” *IEEE International WIE Conference on Electrical and Computer Engineering (WIECON-ECE)*, 2025. [Published — IEEE Xplore]

“**Identification and Translation of Transliterated Chakma Text into Bangla.**” *Undergraduate Thesis — Target: ACL / EMNLP Workshop or Q1/Q2 NLP Journal*, 2026. [Defended — Manuscript in Preparation]

RESEARCH EXPERIENCE

Undergraduate Thesis Researcher

Jan 2025 – May 2026

Dept. of Computer Science & Engineering, MIST, Dhaka, Bangladesh

Supervisor: Lt Col Muhammad Nazrul Islam, PhD • Associate Professor, CSE — MIST • nazrul@cse.mist.ac.bd
Top 2% Scientist globally (Stanford-Elsevier 2022–2025) • 69+ Q1/Q2 journal articles • USD 250,000+ in research grants

- Co-authored two IEEE conference papers published prior to graduation: *Automated Door Lock System Using Face Recognition* (IEEE QPAIN 2025, **Published**) and *Development of a Sustainable Integrated Agricultural IoT System* (IEEE WIECON-ECE 2025, **Published**).
- Completed task-adaptive fine-tuning protocols for Bangla↔Chakma translation using the T5 framework; built BLEU/chrF++ evaluation pipelines and a web deployment interface for Chakma-speaking communities. Thesis successfully defended; manuscript now in preparation for Q1/Q2 journal submission.
- Contributed to experimental design, methodology discussions, and manuscript preparation within an active ML/NLP research group.

RESEARCH & TECHNICAL PROJECTS

Identification and Translation of Transliterated Chakma Text into Bangla [Undergraduate Thesis — Defended]

Jan 2025 – May 2026

Python • PyTorch • Hugging Face • T5 / mT5 • Seq2Seq • BLEU / chrF++ Evaluation

- Fine-tuned a Bengali T5 model for bidirectional Chakma↔Bengali translation, targeting two linguistically distinct low-resource languages with very limited parallel corpora — Chakma is an indigenous language spoken by roughly 400,000 people with almost no digital NLP resources.
- Benchmarked against mBART and mT5 baselines using BLEU, chrF++, and human preference evaluation to measure the transfer-learning gap in this language pair.
- Deployed as a web application to validate end-to-end system performance and give Chakma speakers direct access to the model. Manuscript now in preparation for Q1/Q2 journal submission.

RetailOS — AI-Native Shelf Intelligence & Operations Suite [GitHub]

2026

Next.js • FastAPI • Claude 3.5 • Gemini API • Supabase

- Developed an AI-driven visual analytics platform for real-time retail shelf-compliance monitoring, enabling automated planogram auditing at scale.
- Engineered multi-LLM vision pipelines combining Claude 3.5 and the Gemini API for product detection, recognition, and compliance verification.
- Built a full-stack Next.js / FastAPI architecture with an offline-first sync mechanism and an integrated RetailGPT chatbot for context-aware database querying.
- Implemented a 4-tier fraud-prevention engine — MD5 hash deduplication, image sharpness validation, Haversine-based geofence verification, and timestamp auditing — to flag GPS spoofing, duplicate submissions, and falsified field reports.

Churn Sentinel — Explainable AI Customer Retention System [GitHub]

2026

Python • XGBoost • SHAP • LangChain • Multi-Agent AI

- Trained an XGBoost classifier with stratified cross-validation and Optuna hyperparameter search on class-imbalanced telecom churn data.
- Integrated SHAP for global (beeswarm, summary bar) and local (force plot, waterfall) explanations to make model predictions auditable at the individual decision level.
- Built a LangChain multi-agent pipeline (Behavior Detector → Discount Evaluator → Personalized Outreach Generator) that converts SHAP risk scores into autonomous, context-aware retention actions.

Adaptive Traffic Signal Control via Reinforcement Learning [GitHub]

2025

Python • PPO • Stable-Baselines3 • SUMO Simulator • OpenAI Gym

- Built a custom Gym-compatible SUMO environment with normalized per-lane state features and a composite reward combining queue-length penalty and vehicle-throughput bonus.
- Enforced safe exploration through yellow-phase protection constraints, preventing conflicting signal transitions during agent training.
- Trained a PPO agent for 200,000 steps with EvalCallback checkpointing and TensorBoard logging; the learned policy reduced average queue length relative to a fixed-cycle baseline across all evaluation episodes.

Bayesian Water Quality Analysis — Bangladesh [GitHub]

2025

Python • PyMC • Bayesian Hierarchical Modeling • Environmental Data Science

- Applied Bayesian hierarchical models to quantify uncertainty in water quality measurements across geographically distributed sampling sites in Bangladesh, where chronic arsenic exposure affects an estimated 20 million people.
- Derived posterior predictive distributions for arsenic, dissolved solids, and microbial thresholds, producing probabilistic risk estimates rather than binary pass/fail classifications.
- Dataset and methodology contribute to the evidence base for policy decisions on rural water safety in South Asia.

Genomic Variant Classifier — Pathogenicity Prediction [GitHub]

2025

Python • Scikit-Learn • Random Forest • Bioinformatics

- Built a multi-class Random Forest classifier to predict SNV pathogenicity (benign / pathogenic / variant of uncertain significance) using ClinVar-annotated variants with in-silico features including SIFT, PolyPhen-2, and allele frequency scores.
- Addressed severe class imbalance between pathogenic and benign variants through stratified sampling and threshold tuning; evaluated using precision, recall, F1, and ROC-AUC per class.
- Motivated by the clinical cost of misclassification: a variant incorrectly labeled benign can delay diagnosis in hereditary cancer and cardiac conditions where early intervention is decisive.

PROFESSIONAL EXPERIENCE

Software Engineering Intern — Aurify (Part-time, Remote)

Aug 2025 – Apr 2026

Chicago, IL, USA • Node.js / React SaaS Platform

- Designed Playwright / JavaScript E2E test suites (Page Object Model) across critical user flows and developed a structured defect triage process that systematically reduced regression escapes over three consecutive production release cycles — applying the same hypothesis-driven thinking used in experimental research to software quality problems.
- Diagnosed full-stack bugs in a Node.js / React codebase, validated REST API contracts via Postman, and produced 30+ reproducible defect reports with environment-specific reproduction steps — work that required the same precision and documentation discipline expected in research settings.

TECHNICAL SKILLS

Languages	Python, C++, JavaScript, SQL, LaTeX
ML & Deep Learning	Scikit-Learn, XGBoost, PyTorch, Hugging Face Transformers, T5 / BERT / mT5 Fine-tuning, Stable-Baselines3, LangChain
Explainability	SHAP, LIME, feature attribution, model cards
Probabilistic Methods	Bayesian inference, PyMC, hierarchical modeling, posterior predictive checks
Reinforcement Learning	PPO, DQN, OpenAI Gym, SUMO simulator
Research Tools	Jupyter, MATLAB, Git / GitHub, LaTeX, Overleaf, Weights & Biases
Software Engineering	FastAPI, React, Node.js, Playwright, PostgreSQL, REST APIs

COMPETITIVE PROGRAMMING

300+ problems solved across LeetCode, Codeforces, and CodeChef, covering data structures, graph algorithms, dynamic programming, and combinatorics.

LeetCode: rubayetkabirzisan • Codeforces: mission27 • CodeChef: rubaet_kabir

STANDARDIZED TESTS

GRE General Test	Preparation in progress; examination to be scheduled.
IELTS Academic	Preparation in progress; examination to be scheduled.

REFERENCES

Lt Col Muhammad Nazrul Islam, PhD	Dr. Md. Akhtaruzzaman	Dr. Fernaz Narin Nur
Associate Professor, CSE Military Institute of Science & Technology (MIST) nazrul@cse.mist.ac.bd	Associate Professor & Head Dept. of Robotics & Mechatronics Engineering, DIU akhtaruzzaman.cse@diu.edu.bd	Professor, CSE Daffodil International University Dhaka, Bangladesh fernaz.cse@diu.edu.bd
<i>Thesis supervisor.</i>	<i>Co-author, IEEE QPAIN 2025.</i>	<i>Co-author, IEEE WIECON 2025.</i>
<i>Top 2% Scientist</i>	<i>PhD, IIUM Malaysia.</i>	<i>PhD, Univ. of Dhaka.</i>
<i>(Stanford-Elsevier 2022–25).</i>	<i>Formerly, MIST CSE.</i>	<i>Formerly, MIST CSE.</i>